while

$$\jmath(\mathcal{Q}) > 0 \text{ implies } \begin{cases} \chi_H(\mathcal{S}_n) = \chi_H(\mathcal{Q}) < 1, \\ \mathcal{D}_{1,1}(\mathcal{S}^\circ) \text{ is satisfied with } \varepsilon(r) = \hbar(\mathcal{Q}) > 0 \text{ for any } r \geqslant 1, \text{ and} \\ \sup_{n\geqslant 0} \chi_\Phi(\mathcal{S}_n) \leqslant \chi(\mathcal{S}) := \sup_{n\geqslant 0} \chi(\mathcal{S}_n) \leqslant 1 - \hbar(\mathcal{Q}) < 1. \end{cases} \tag{4.2}$$

We can also obtain the sandwich-type contraction inequalities described below.

Theorem 4.1. *For any convex function Φ satisfying (1.12) we have the contraction estimates*

$$\begin{aligned} D_\Phi\left(\pi_{2(n+1)}, \nu_V\right) &\leqslant \chi(\mathcal{S})\ D_\Phi\left(\lambda_U, \pi_{2n+1}\right) \leqslant \chi(\mathcal{S})^2\ D_\Phi\left(\pi_{2n}, \nu_V\right), \\ D_\Phi\left(\nu_V, \pi_{2(n+1)}\right) &\leqslant \chi(\mathcal{S})\ D_\Phi\left(\pi_{2n+1}, \lambda_U\right) \leqslant \chi(\mathcal{S})^2\ D_\Phi\left(\nu_V, \pi_{2n}\right). \end{aligned} \tag{4.3}$$

The above inequalities remain valid when we replace (ν_V, π_{2n}) by (λ_U, π_{2n+1}). If we assume that $\jmath(\mathcal{Q}) > 0$, then $\chi(\mathcal{S}) < 1$ and, in addition, there exists some $c > 0$ such that

$$\left\| \log \frac{d\pi_{2n}}{d\nu_V} \right\| \vee \left\| \log \frac{d\pi_{2n+1}}{d\lambda_U} \right\| \leqslant c\ \chi(\mathcal{S})^{2n} \quad \textit{for any } n \geqslant 0. \tag{4.4}$$

Proof. The estimates (4.3) are a direct consequence of (3.17) combined with the Markov transport formulae

$$\begin{aligned} (\pi_{2(n+1)}, \nu_V) &= (\lambda_U \mathcal{S}_{2(n+1)}, \pi_{2n+1}\mathcal{S}_{2(n+1)}), \quad \text{and} \\ (\lambda_U, \pi_{2n+1}) &= (\pi_{2n}\mathcal{S}_{2n+1}, \nu_V \mathcal{S}_{2n+1}), \end{aligned}$$

which are themselves easily obtained from expressions (2.2) and (2.8). The proof of (4.4) follows exactly the same arguments as the proof of the inequality (4.24) in Theorem 4.8, hence it is skipped. ∎

The following result can be obtained from the series expansion given by Corollary 4.15.

Corollary 4.2. *If we assume that $\jmath(\mathcal{Q}) > 0$, then $\chi(\mathcal{S}) < 1$ and, in addition, there exists some $c > 0$ such that*

$$\mathrm{Ent}(\mathbb{P} \mid \mathcal{P}_{2n}) \leqslant c\ \chi(\mathcal{S})^{2n}, \quad \textit{for any } n \geqslant 0.$$

Several examples satisfying condition $\jmath(\mathcal{Q}) > 0$ are discussed in Section 2.4 in [40]. For instance, consider a parameter $\delta > 0$ and drift function $x \mapsto b(x)$ with bounded oscillations from $\mathbb{X} = \mathbb{R}^d$ into itself. The choice $\mathcal{Q}(x, dy) \propto e^{-\delta\,\|y - b(x)\|}\ dx$ yields

$$\jmath(\mathcal{Q}) \geqslant \exp\left(-\delta\,\Delta(b)\right), \quad \text{with} \quad \Delta(b) = \sup_{(x_1,x_2)\in\mathbb{X}^2} \|b(x_1) - b(x_2)\|.$$

In continuous time settings, condition $\jmath(\mathcal{Q}) > 0$ is also satisfied for Markov transitions of elliptic diffusions on compact manifolds $\mathbb{X}$ –see for instance the pioneering work of Aronson [7], Nash [84] and Varopoulos [103] on Gaussian estimates for heat kernels on manifolds. Nevertheless, the above condition is not met for Gaussian transitions.

4.2. **Gibbs-Loop Sinkhorn semigroups.** For a positive function $\varphi \in \mathcal{B}(\mathbb{X})$ and $r > 0$, we define its r-sublevel set by

$$C_\varphi(r, \mathbb{X}) := \{x \in \mathbb{X}\ :\ \varphi(x) \leqslant r\}. \tag{4.5}$$

When there is no ambiguity, we write $C_\varphi(r) = \{\varphi \leqslant r\}$.

Lyapunov functions combined with local minorization conditions are classical tools in the analysis of time-inhomogeneous Markov chains; see, for instance, [6] and the references therein.

In the context of Gibbs-loop semigroups (2.9), such conditions are formulated using Lyapunov-type functions $\varphi \in \mathcal{B}(\mathbb{X})$ and $\psi \in \mathcal{B}(\mathbb{Y})$ satisfying $\varphi \wedge \psi \geqslant 1$, as described below.

Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$: *There exist $\epsilon \in (0,1)$, a constant $c > 0$, and an integer $n_0 \geqslant 1$ such that $\pi_{2n_0}(\psi) \vee \pi_{2n_0+1}(\varphi) < \infty$ and, for all $n \geqslant n_0$,*

$$\mathcal{S}^\circ_{2n}(\psi) \leqslant \epsilon, \psi + c, \qquad \mathcal{S}^\circ_{2n+1}(\varphi) \leqslant \epsilon, \varphi + c. \tag{4.6}$$

Moreover, there exist $r_0 \geqslant 1$ and a function

$$\varepsilon : [r_0, \infty[\to]0, 1]$$

such that, for any $r \geqslant r_0$, $(x_1, x_2) \in C_\varphi(r)^2$, $(y_1, y_2) \in C_\psi(r)^2$, and $n \geqslant n_0$,

$$\|(\delta_{x_1} - \delta_{x_2})\mathcal{S}^\circ_{2n+1}\|_{\mathsf{tv}} \vee \|(\delta_{y_1} - \delta_{y_2})\mathcal{S}^\circ_{2n}\|_{\mathsf{tv}} \leqslant 1 - \varepsilon(r). \tag{4.7}$$

Condition (4.7) holds if and only if there exist probability measures $\mathcal{X} \in \mathcal{M}_1(\mathbb{X})$ and $\mathcal{Y} \in \mathcal{M}_1(\mathbb{Y})$, possibly depending on $(n, r, (x_1, x_2))$ and $(n, r, (y_1, y_2))$, respectively, such that, for $i \in 1, 2$,

$$\mathcal{S}^\circ_{2n}(y_i, dy) \geqslant \varepsilon(r)\ \mathcal{Y}(dy) \quad \text{and} \quad \mathcal{S}^\circ_{2n+1}(x_i, dx) \geqslant \varepsilon(r)\ \mathcal{X}(dx). \tag{4.8}$$

The drift condition (4.6) is trivially satisfied for the constant functions $\varphi = 1 = \psi$. In this case, $C_\varphi(r) = \mathbb{X}$ and $C_\psi(r) = \mathbb{Y}$ for all $r \geqslant 1$, and (4.8) reduces to a uniform minorization condition over the whole state spaces.

Let $\mathcal{B}_\infty(\mathbb{X}) \subset \mathcal{B}(\mathbb{X})$ denote the subalgebra of locally bounded, uniformly positive functions φ that grow at infinity, in the sense that $\sup_C \varphi < \infty$ for any compact set $C \subset \mathbb{X}$ and that, for any

$$r \geqslant \varphi_\star := \inf_{\mathbb{X}} \varphi > 0, \tag{4.9}$$

the sublevel set $C_\varphi(r)$ is nonempty and compact. We further define

$$\mathcal{B}_0(\mathbb{X}) := \{1/\varphi \ :\ \varphi \in \mathcal{B}_\infty(\mathbb{X})\} \subset \mathcal{B}_b(\mathbb{X})$$

the algebra of bounded, positive functions that are locally lower bounded and vanish at infinity. If $\varphi \in \mathcal{B}_\infty(\mathbb{X})$ and $\psi \in \mathcal{B}_\infty(\mathbb{Y})$, then the sets $C_\varphi(r)$ and $C_\psi(r)$ are compact, and (4.8) yields genuine local minorization conditions.

The assumption $\varphi \wedge \psi \geqslant 1$ is made solely for notational convenience. Indeed, if $\varphi \in \mathcal{B}_\infty(\mathbb{X})$ and $\psi \in \mathcal{B}_\infty(\mathbb{Y})$ satisfy (4.6), then the rescaled functions $\varphi/\varphi_\star$ and $\psi/\psi_\star$ also satisfy the same drift inequalities.

The drift conditions (4.6) imply that, for all $n \geqslant n_0$,

$$\mathcal{S}^\circ_{2n}\left(\mathcal{B}_\psi(\mathbb{Y})\right) \subset \mathcal{B}_\psi(\mathbb{Y}) \quad \text{and} \quad \mathcal{S}^\circ_{2n+1}\left(\mathcal{B}_\varphi(\mathbb{X})\right) \subset \mathcal{B}_\varphi(\mathbb{X}).$$

Moreover, whenever $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ holds,

$$\lambda_U(\varphi) \vee \nu_V(\psi) < \infty \quad \text{and} \quad \sup_{n \geqslant n_0} \left(\pi_{2n}(\psi) \vee \pi_{2n+1}(\varphi)\right) < \infty. \tag{4.10}$$

These bounds follow directly from the drift inequalities (4.6); a detailed proof is given in Section E.

4.3. **Weighted norm contractions.** Combining the drift conditions (4.6) with the local minorization conditions (4.7) allows us to control the contraction coefficients of the Sinkhorn transitions. More precisely, we obtain the following uniform contraction estimate for the forward–backward Markov kernels $\mathcal{S}^\circ_{2n}$ and $\mathcal{S}^\circ_{2n+1}$.

Theorem 4.3. *Assume that Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ holds for some $n_0 \geqslant 1$. Then there exist parameters $\kappa > 0$ and $\rho \in [0,1[$ depending only on (ϵ, c) in (4.6) and on $\varepsilon(r)$ in (4.7) and such that*

$$\sup_{n \geqslant n_0} \left(\chi_{\psi_\kappa}(\mathcal{S}^\circ_{2n}) \vee \chi_{\varphi_\kappa}(\mathcal{S}^\circ_{2n+1})\right) \leqslant \rho, \quad \text{where} \quad (\varphi_\kappa, \psi_\kappa) := (1+\kappa\varphi, 1+\kappa\psi). \tag{4.11}$$

Theorem 4.3 is an immediate consequence of Theorem C.1 in Appendix C.

The contraction estimate (4.11) ensures the exponential decay of Sinkhorn semigroups. For instance, for any $\mu_1, \mu_2 \in \mathcal{M}_{1,\psi}(\mathbb{Y})$ and $n_0 \leqslant p \leqslant n$ we readily obtain

$$\left\|\!\left|(\mu_1 - \mu_2)\mathcal{S}^\circ_{2(p-1),2n}\right|\!\right\|_{\psi_\kappa} \leqslant \rho^{1+n-p} \; |\!|\!|\mu_1 - \mu_2|\!|\!|_{\psi_\kappa}.$$

Note that

$$\kappa\, \psi \leqslant \psi_\kappa \leqslant (1+\kappa)\psi \quad \text{implies} \quad \left\|\!\left|(\mu_1 - \mu_2)\mathcal{S}^\circ_{2(p-1),2n}\right|\!\right\|_{\psi} \leqslant (1+1/\kappa)\; \rho^{1+n-p} \; |\!|\!|\mu_1 - \mu_2|\!|\!|_{\psi}.$$

Choosing $p = n_0$ and $(\mu_1, \mu_2) = (\pi_{2(n_0-1)}, \nu_V)$, the above estimate immediately yields, for any $n \geqslant n_0$, an upper bound on the ψ-norm. We summarize the above discussion with the following corollary.

Corollary 4.4. *Under the assumptions of Theorem 4.3 for any $n \geqslant n_0$ we have*

$$\begin{aligned} |\!|\!|\pi_{2n} - \nu_V|\!|\!|_{\psi} &\leqslant (1+1/\kappa)\; \rho^{1+n-n_0} \; \left\|\!\left|\pi_{2(n_0-1)} - \nu_V\right|\!\right\|_{\psi} \\ |\!|\!|\pi_{2n+1} - \lambda_U|\!|\!|_{\varphi} &\leqslant (1+1/\kappa)\; \rho^{1+n-n_0} \; |\!|\!|\pi_{2n_0-1} - \lambda_U|\!|\!|_{\varphi}, \end{aligned} \tag{4.12}$$

with the parameters $\rho \in [0,1[$ and $\kappa > 0$ as in (4.11).

Choosing $(\mu_1, \mu_2) = (\delta_{y_1}, \delta_{y_2})$ for some $(y_1, y_2) \in \mathbb{Y}^2$, we can also verify the merging property of Sinkhorn semigroups [35, 94, 95]

$$\|\mathcal{S}^\circ_{2(n_0-1),2n}(y_1, .) - \mathcal{S}^\circ_{2(n_0-1),2n}(y_2, .)\|_{\mathsf{tv}} \leqslant \rho^{1+n-n_0} \; \left(\psi_\kappa(y_1) + \psi_\kappa(y_2)\right).$$

Remark 4.5. *The proximal sampler discussed in Remark 2.1 has a Markov transition $\mathcal{S}^\circ_1$ with invariant measure λ_U. The weighted-norm contraction properties of these models are simply based on the existence of a Lyapunov function φ satisfying the r.h.s. condition in (4.6) with $n = 0$. As in (4.7) we also require a local minorization condition $\|(\delta_{x_1} - \delta_{x_2})\mathcal{S}^\circ_1\|_{\mathsf{tv}} \leqslant 1 - \varepsilon(r)$ on the sub-level sets $(x_1, x_2) \in C_\varphi(r)^2$ of the Lyapunov function to ensure that $\chi_{\varphi_\kappa}(\mathcal{S}^\circ_1) < 1$, for some parameter $\kappa \geqslant 0$ (see for instance Theorem C.1).*

Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is expressed in terms of the forward-backward transition kernels $\mathcal{S}^\circ_{2n}$ and $\mathcal{S}^\circ_{2n+1}$, which may not be easy to test – except in the case of Gaussian and linear models. Several Lyapunov functions in this context are designed in Section 5.8. A stronger (but easier to test) condition can be provided in terms of the transitions $\mathcal{S}_{2n}$ and $\mathcal{S}_{2n+1}$.

Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ *There exists $\epsilon \in]0,1[$, a constant $c > 0$ and an integer $n_0 \geqslant 0$ such that for any $n \geqslant n_0$,*

$$\mathcal{S}_{2n}(\psi) \leqslant \epsilon\; \varphi + c, \qquad \mathcal{S}_{2n+1}(\varphi) \leqslant \epsilon\; \psi + c. \tag{4.13}$$

Moreover, there exists some $r_0 \geqslant 1$ and a function

$$\varepsilon : [r_0, \infty[\to]0,1]$$

such that for any $r \geqslant r_0$, $(x_1, x_2) \in C_\varphi(r)^2$, $(y_1, y_2) \in C_\psi(r)^2$ and $n \geqslant n_0$,

$$\|(\delta_{x_1} - \delta_{x_2})\mathcal{S}_{2n}\|_{\mathsf{tv}} \vee \|(\delta_{y_1} - \delta_{y_2})\mathcal{S}_{2n+1}\|_{\mathsf{tv}} \leqslant 1 - \varepsilon(r). \tag{4.14}$$

When Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is met, for any $n \geqslant n_0$ we readily see that

$$\mathcal{S}_{2n}\left(\mathcal{B}_\psi(\mathbb{Y})\right) \subset \mathcal{B}_\varphi(\mathbb{X}) \quad \text{and} \quad \mathcal{S}_{2n+1}\left(\mathcal{B}_\varphi(\mathbb{X})\right) \subset \mathcal{B}_\psi(\mathbb{Y}).$$